
Haptic Pucks & Pollen Pods

**Complete Variant Reference — All Iterations · BOM Consolidation ·
Fabrication & Deployment Reference**

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Section 1 — Design Ecosystem Overview

The Wrist-Puck and Sensor Species Pollen Pod families are distinct morphologies sharing a common fabrication and electronics vocabulary. Both are 3D-printed enclosures housing sensors and/or actuators. Both route I2C buses internally, drive

addressable WS2812B LED arrays, and operate as nodes in a larger distributed network coordinated over Wi-Fi. Both are designed and built by Sean Pond, Whitehorse, YT, Canada, from a Canada-sourced BOM with remote shipping constraints baked into the budget.

The **Wrist-Puck** is a wearable haptic output device. It receives JSON motifs over UDP from a sending application and translates them into dual-channel haptic feedback: a DRV2605L-driven linear resonant actuator (LRA) for texture and vibrotactile sensation, and an SG51R sub-micro servo driving a sliding brass mass for weight-shift and momentum cues. It is a self-contained, battery-powered node strapped to the wrist. Its microcontroller is a Seeed XIAO ESP32-C3.

The **Sensor Species Family** is a fixed environmental sensing ecosystem composed of four named morphologies: Pollen Pod (sensor capsule), Perch Pod (sentinel/motion detector), Reef Plate (structural mounting substrate), and Garland Links (kinetic connective tissue). Together they form an articulated, illuminated, sensor-rich installation network. Node compute runs on Raspberry Pi Pico W modules reporting to a Raspberry Pi 4B hub. Sensors span temperature, humidity, barometric pressure, high-dynamic-range light, gesture, proximity, PIR motion, and capacitive touch. LED choreography is broadcast from the central hub across WS2812B strips and NeoPixel arrays woven through the garland.

The platforms are intentionally kept architecturally separate — the Wrist-Puck is standalone; the Sensor Species family is networked infrastructure — but they share: JST connector vocabulary (JST-SH for I2C daisy-chain, JST-PH for power), WS2812B as the common LED language, PETG/TPU material conventions, and a design philosophy favouring parametric OpenSCAD enclosures with precise fit tolerances, serviceable hand-crimped connections, and modular scalability.

This document is structured so that each section stands alone as a self-contained lookup. Cross-references are noted where relevant.

Platform Comparison Table

Platform	Role	Form Factor	Microcontroller	Key Actuators / Sensors	Comms Protocol	Power	Enclosure Material	Status
Wrist-Puck (v1.0)	Wearable haptic output	68 mm dia × 22 mm puck	Seeed XIAO ESP32-C3	DRV2605L LRA + SG51R servo (sliding mass)	Wi-Fi UDP port 9000 (JSON motifs)	3.7 V 500 mAh LiPo, USB-C charge	PLA or PETG	Rev 1.0 active
Pollen Pod (Species A)	Fixed environmental sensor capsule	Ovoid clamshell, ~80–100 mm longest.	Raspberry Pi Pico W	BME280 + TSL2591 + capacitive touch + WS2812B LED ring	Wi-Fi to RPi 4B hub	5 V main PSU + LiPo backup	PETG (shell) + TPU 95A (vent ring) + Clear PETG (lens)	Active, v1 build campaign
Perch Pod (Species B)	Sentinel motion/gesture detector platform	Nest form, larger than Pollen Pod	Raspberry Pi Pico W	PIR motion + APDS-9960 gesture/proximity + NeoPixel Jewel	Wi-Fi to RPi 4B hub	5 V main PSU + LiPo backup	PETG + Clear PETG dome	Active, v1 build campaign
Reef Plate (Species C)	Structural mounting substrate	200 × 150 mm backplane panels	No MCU (passive structure)	None (mounting surface only)	N/A	N/A	PETG (primary) / ASA (UV-stable variant)	Active, v1 build campaign
Garland Links (Species D)	Kinetic connective tissue	Chain link forms, ~48 links + 8 bloom nodes + 16 drift fins	WS2812B LED strip shared with pod nodes	None dedicated	Shared LED data line	Shared PSU	PETG (links) + TPU spacer rings + PETG-CF (drift fins)	Active, v1 build campaign

Section 2 — Wrist-Puck Haptic Node (v1.0)

Self-contained wearable haptic output node. Dual-channel haptic: LRA for texture/vibrotactile + servo-driven sliding mass for weight-shift. Receives JSON motifs over Wi-Fi UDP. Designed for solo wrist wear or multi-node deployment (future ESP-NOW mesh). All specifications below are for v1.0.

2.1 Key Specifications Table

Parameter	Value
Puck diameter	68 mm
Height	22 mm
Weight (assembled, est.)	~55 g
Wireless	Wi-Fi 802.11 b/g/n, BLE 5.0
Protocol	UDP port 9000, JSON motifs
Battery	3.7 V, 400–500 mAh LiPo
Runtime (est.)	2–4 hours active use
Charging	USB-C via Adafruit PowerBoost 500C (500 mA charge current, single-board charge + boost)
Power management board	Adafruit PowerBoost 500C — 3.7 V LiPo → 5.2 V regulated output, 500–1000 mA, LBO low-battery pin
Haptic channel 1	10 mm coin LRA driven by DRV2605L (texture / vibrotactile)
Haptic channel 2	Adafruit SG51R sub-micro servo + sliding brass mass ~8–10 g (weight-shift)
Enclosure	3D printed, PLA or PETG, 0.2 mm layer / 20% infill
Wrist attachment	22 mm NATO or silicone watch strap

2.2 Bill of Materials — Wrist-Puck v1.0 (Canada-Sourced, CAD, May 2026)

#	Compon ent	Descrip tion	Supplie r	SKU / ASIN	Qty	Est. Price (CAD)	Notes
1	ESP32-C3 Dev Board	Seeed Studio XIAO ESP32-C3, 21×17.8 mm, Wi-Fi+BLE, USB-C, LiPo charging built-in	Amazon.ca	ASIN: B0B94JZ2YF	1	~\$43.73	Amazon's Choice; ships free. Built-in LiPo charge circuit makes TP4056 optional.
2	Haptic Driver	DRV260 5L I2C Haptic Motor Controller Breakout Board (3-5 V)	Amazon.ca	Search "DRV260 5L breakout" (NUEAY MS or similar)	1	~\$13.81	Generic breakout compatible with Adafruit library. Verify I2C pull-ups present on board.
3	10 mm Coin LRA	10 mm Linear Resonant Actuator, C10-100 type, ~175 Hz, 2.0 V rated	DigiKey.ca	C10-100 (Precision Microdrives) or equiv.	1	~\$8-12	Substitute with any 8-12 mm coin LRA. Check resonant frequency for DRV260 5L auto-cal.
4	Sub-Micro Servo	Adafruit SG51R, 5 g, 25×22×12 mm, 0.6 kg·cm torque, 4.8 V	Mouser.ca	Part #: 485-2201	1	~\$8.63	Not readily on Amazon.ca. Alt: PiShop.ca (~\$12.95). Substitute: any 5-9 g micro servo.
5	LiPo	3.7 V	Amazon.	Search	1	~\$20.00	Verify

#	Component	Description	Supplier	SKU / ASIN	Qty	Est. Price (CAD)	Notes
	Battery	500 mAh Lithium Polymer, JST-PH 2.0 connector, ~35×25×5 mm	ca	"EEMB 500mAh 602040 LiPo JST"			JST polarity matches XIAO before connecting! XIAO uses JST-SH 1.0 mm natively; may need adapter or direct solder.
6	Power Management Board	Adafruit PowerBoost 500C — LiPo boost to 5.2 V regulated, 500–1000 mA output, 500 mA LiPo charge, LBO low-battery pin, USB-C input	Adafruit / PiShop.ca	Adafruit #1944	1	~\$14.95 USD (~\$20 CAD est.)	IN HAND — already acquired. Single board replaces TP4056 + voltage divider. Powers XIAO + servo simultaneously at rated voltage.
7	JST Connector Kit	JST PH 2.0 mm Wire Connector Kit, 450 pcs, 2/3/4-pin housings + female headers	Amazon.ca	Search "ZYNCUE JST PH 2.0mm 450pcs"	1 kit	~\$10.79	For battery, servo, LRA, and inter-board wiring. Enough for multiple nodes.
8	Perfboard	Small prototype PCB board, ~20×15	Amazon.ca	Search "mini perfboard assortm"	1	~\$1.50	From assortment pack (~\$10–12 for

#	Component	Description	Supplier	SKU / ASIN	Qty	Est. Price (CAD)	Notes
		mm or 30×20 mm		ent"			30+ boards). Used for wire junction and I2C pull-up resistors.
9	Resistors	4.7 kΩ ×2 (I2C pull-ups, if not on DRV board)	Amazon.ca	From any resistor assortment kit	1 kit	~\$0.50	100 kΩ ×2 no longer needed for voltage divider (replaced by PowerBoost LBO pin). Keep 4.7 kΩ ×2 for I2C pull-ups if not present on DRV breakout. 1/4 W or 1/8 W through-hole.
10	Hookup Wire	26–30 AWG silicone stranded wire, assorted colors	Amazon.ca	Search "30 AWG silicone wire kit"	1	~\$12.00	Flexible silicone insulation preferred for wearable routing.
11	Sliding Mass	Brass or steel slug, ~8–10 g, ~5×5×12 mm (or M5 brass hex nut stacked)	Hardware store / Amazon.ca	Search "M5 brass nut" or machine a small slug	1	~\$2.00	Rides in the 6 mm-wide rail slot alongside the servo. Attach to servo horn via short linkage

#	Compon ent	Descrip tion	Supplie r	SKU / ASIN	Qty	Est. Price (CAD)	Notes
							arm.
12	M2 Hardw are	M2×6 mm screws + M2 nuts (for mountin g boards to enclosur e posts)	Amazon. ca	Search "M2 screw nut assortm ent"	4 sets	~\$0.50	Nylon preferre d to avoid shorts.
13	3D Print Filame nt	PLA or PETG, any color, ~25 g per puck	User's stock	N/A	~25 g	~\$1.00. 0.2 mm layer height, 20% infill recommended.	
14	Wrist Strap	22 mm watch band (NATO style or silicone sport band)	Amazon. ca	Search "22mm NATO watch strap"	1	~\$8-15	Threads through the enclosur e strap slots.

BOM Total

~\$115-135 CAD per node (PowerBoost 500C already in hand; May 2026, Canada-sourced). Excludes Whitehorse shipping surcharge (~10-15% on specialty components from Mouser / DigiKey). For multi-node builds, kits (resistors, JST, perfboard) amortize across multiple units.

2.3 Wiring & Pin Assignment Table

Signal	XIAO ESP32- C3 Pin	Target	Wire Color	Notes
I2C SDA	GPIO6 (D4)	DRV2605L SDA	Blue	4.7 kΩ pull-up to 3.3 V if

Signal	XIAO ESP32-C3 Pin	Target	Wire Color	Notes
				needed
I2C SCL	GPIO7 (D5)	DRV2605L SCL	Yellow	4.7 kΩ pull-up to 3.3 V if needed
Servo PWM	GPIO5 (D3)	SG51R Signal (Orange)	Orange	Direct connection
Low Battery	GPIO2 (D0)	PowerBoost LBO pin	Green	Active LOW when LiPo < 3.2 V — clean binary signal, replaces voltage divider
3.3 V Power	3V3	DRV2605L VIN	Red	3.3 V from XIAO onboard reg
5.2 V Power	5V pin	From PowerBoost 5Vo	Red	PowerBoost feeds XIAO 5V pin — powers XIAO + servo simultaneously
Servo VCC	—	PowerBoost 5Vo → SG51R Red	Red	5.2 V — servo runs at rated voltage, problem closed
GND	GND	All GND	Black	Common ground bus through PowerBoost
LRA+	—	DRV2605L OUT+	White	DRV drives LRA directly
LRA-	—	DRV2605L OUT-	White	DRV drives LRA directly
Battery+	PowerBoost BAT+	LiPo + (JST)	Red	PowerBoost is sole charge + boost manager
Battery-	PowerBoost BAT-	LiPo -	Black	Common ground
USB-C (charge)	PowerBoost USB	Wall / PC USB	—	Charging flows through PowerBoost, not XIAO USB

⚠ **Critical — JST Polarity Warning**

Verify LiPo JST polarity against PowerBoost BAT input **before first connection — multimeter first, always**. Reversed polarity **WILL destroy the board instantly** and may cause a LiPo fire. Adafruit boards are wired for Adafruit batteries; third-party cells vary. No exception.

Power Architecture Note — PowerBoost 500C

The PowerBoost 500C is the single power management layer. LiPo connects to PowerBoost BAT. USB-C charges through PowerBoost. The regulated 5.2 V output feeds both XIAO (via 5V pin) and the servo — servo now runs at rated voltage on battery, eliminating the boost converter fork previously noted in v1.0. XIAO generates its own 3.3 V rail from the 5.2 V input for DRV2605L. The LBO pin provides a clean active-LOW signal when LiPo drops below 3.2 V — simpler and more reliable than the 100k/100k voltage divider approach.

2.4 Assembly Sequence Summary

1. **Print enclosure** — Bottom shell + lid, OpenSCAD parametric file, 0.2 mm layer height, 20% infill, PLA or PETG. Estimated print time: ~1.5 hours per unit. Verify dimensional fit before populating.
2. **Prepare perfboard junction** — Populate power bus, voltage divider (100k/100k from VBAT to GND, midpoint to GPIO2), and I2C pull-up resistors (4.7 k Ω SDA/SCL to 3.3 V, if not already on DRV2605L breakout board). Test connections with continuity before next step.
3. **Wire DRV2605L** — VIN to 3.3 V, GND to common bus, SDA to GPIO6, SCL to GPIO7. LRA leads to OUT+ and OUT-. Verify I2C address 0x5A responds in firmware before enclosure integration.
4. **Wire SG51R servo** — 3-pin connector: Brown=GND, Red=VCC (5 V USB rail), Orange=Signal (GPIO5). Trim leads to ~60 mm for enclosure routing.

Attach smallest servo horn provided. Confirm sweep 750–2250 μ s, center 1500 μ s in firmware.

5. **Prepare sliding mass assembly** — M5 brass hex nuts stacked or machined slug (~8–10 g). Drill 1 mm hole for linkage wire. Verify free movement in 6 mm × 30 mm rail slot without binding. Attach to servo horn via short linkage arm — verify full sweep does not exceed rail travel.
6. **Install components in enclosure** — Sequence: (a) LRA: press-fit face-down into formed pocket, CA glue if needed. (b) Servo: M2 screws or friction fit into servo bay. (c) Sliding mass + linkage: confirm articulation. (d) DRV2605L: foam tape or M2 standoffs. (e) XIAO ESP32-C3: USB-C port aligned to enclosure wall cutout. (f) LiPo: foam tape — **DO NOT compress**. (g) TP4056 optional — mount adjacent to XIAO if used. (h) Perfboard on top of battery layer. Route and dress all wires. Snap lid closed — confirm no pinched wires.
7. **Flash firmware** — Arduino IDE 2.x. Board: XIAO_ESP32C3. Libraries required: ESP32Servo, ArduinoJson v7, Wire. Set WIFI_SSID and WIFI_PASS in firmware. Flash and monitor at 115200 baud. Confirm I2C device found at 0x5A, servo sweep test, LRA effect test, Wi-Fi connection, and UDP listener active on port 9000 before closing enclosure.

2.5 JSON Motif Schema

Motifs are JSON objects sent as UDP datagrams to port 9000 on the Wrist-Puck's IP address. The firmware parses and executes steps sequentially. Top-level fields:

Field	Type	Range / Values	Description
id	string	Any string	Motif identifier for logging and dedup
repeat	int	1–10	Number of times to execute the full step array
steps	array	—	Ordered array of step objects (see below)

LRA Step Types:

Step Type	Field	Range	Description
effect	effect	1-123	DRV2605L built-in waveform library effect number
rtp	amplitude	0-255	Real-time playback amplitude (0=off, 255=max)
stop	—	—	Stop LRA immediately
<i>Common</i>	dur_ms	0-5000	Duration before proceeding to next step (ms)

Servo Step Types:

Step Type	Field	Range	Description
pos	pos_us	750-2250 μ s	Move servo to specified pulse width (μ s)
center	—	—	Return servo to 1500 μ s center position
sweep	—	—	Sweep servo full range
<i>Common</i>	dur_ms	0-5000	Duration before proceeding to next step (ms)

Example Motifs:

```
// rough_bump — abrasive texture cue with weight shift {  "id": "rough_bump",
"repeat": 1,  "steps": [  {"type": "lra", "mode": "effect", "effect": 47, "dur_ms": 150},
{"type": "servo", "mode": "pos", "pos_us": 1100, "dur_ms": 200},  {"type": "lra",
"mode": "effect", "effect": 52, "dur_ms": 150},  {"type": "servo", "mode": "center",
"dur_ms": 200}  ] } // weight_left — smooth lateral weight-shift with low rumble
{  "id": "weight_left",  "repeat": 1,  "steps": [  {"type": "servo", "mode": "pos",
"pos_us": 800, "dur_ms": 300},  {"type": "lra", "mode": "rtp", "amplitude": 60,
```

```
"dur_ms": 400},    {"type": "lra", "mode": "stop", "dur_ms": 50},    {"type": "servo",  
"mode": "center", "dur_ms": 300}  ] } // heartbeat — double-pulse cardiac  
sensation, 3 repetitions {  "id": "heartbeat",  "repeat": 3,  "steps": [    {"type": "lra",  
"mode": "effect", "effect": 1, "dur_ms": 100},    {"type": "lra", "mode": "effect",  
"effect": 1, "dur_ms": 150},    {"type": "servo", "mode": "pos", "pos_us": 1350,  
"dur_ms": 200},    {"type": "servo", "mode": "center", "dur_ms": 350}  ] }
```

2.6 Testing Checklist Summary

35 total test items across 7 groups. Complete all pre-power checks before first power-on.

Pre-Power Checks (6 items)

8. Confirm all wiring connections match pin assignment table — no bare wire contacts touching enclosure or each other
9. Verify JST connector polarity on LiPo before connecting — + on left (USB faces you)
10. Measure voltage divider midpoint with multimeter — should read $\sim V_{BAT}/2$ when battery connected
11. Verify I2C pull-up resistors present (on DRV board or on perfboard) — 4.7 k Ω SDA and SCL to 3.3 V
12. Confirm LRA is press-fit flush, face down, with no adhesive bridging the LRA terminals
13. Confirm sliding mass moves freely in rail slot through full servo sweep range — no binding

Power-On Checks (5 items)

14. Connect USB-C — confirm XIAO power LED illuminates, no smoke or heat
15. Open Serial Monitor at 115200 baud — confirm firmware boot messages, no crash loop

16. Confirm DRV2605L detected at I2C address 0x5A in serial output
17. Confirm battery voltage reading plausible (3.3–4.2 V range) on ADC GPIO2
18. Confirm no hardware watchdog resets during boot sequence

Calibration Checks (5 items)

19. Run DRV2605L auto-calibration routine — confirm success flag returned via serial
20. Confirm LRA resonant frequency detected within range for C10-100 (~175 Hz)
21. Sweep servo 750–2250 μ s — confirm full travel without binding or stall
22. Confirm center position (1500 μ s) returns mass to geometric center of rail slot
23. Play DRV2605L effect 47 manually — confirm LRA vibration felt through enclosure surface

Wi-Fi Checks (3 items)

24. Confirm SSID and password correct, 2.4 GHz band (not 5 GHz — ESP32-C3 does not support 5 GHz)
25. Confirm IP address assigned and printed to serial — note for UDP sender configuration
26. Confirm UDP listener active on port 9000 — send test ping from sender host

UDP Motif Tests (6 items)

27. Send "rough_bump" motif — confirm LRA effect 47 fires, servo moves to 1100 μ s, LRA effect 52 fires, servo returns center
28. Send "weight_left" motif — confirm servo moves to 800 μ s, low RTP amplitude felt, LRA stops, servo returns center
29. Send "heartbeat" motif (repeat:3) — confirm double pulse $\times$ 3 repetitions with servo position shift
30. Send malformed JSON — confirm firmware discards gracefully without crash

- 31. Send motif with unknown id — confirm firmware executes steps normally (id is for logging only)
- 32. Send motif with repeat:10 and 4 steps — confirm all 40 step executions complete without watchdog reset

Safety Checks (4 items)

- 33. Confirm idle timeout — after 5 seconds of no motifs received, servo auto-centers
- 34. Confirm hardware watchdog — introduce >8 s blocking delay in test firmware; confirm reset and recovery
- 35. Confirm low battery alert — with battery discharged below 3.3 V, confirm motors stop and alert prints to serial
- 36. Confirm LiPo does not become warm after 30 minutes of active use — thermal check of enclosure exterior

Enclosure Fit Checks (6 items)

- 37. Confirm lid snaps flush with no component interference inside
- 38. Confirm USB-C port accessible through wall cutout with connector — no forced angle
- 39. Confirm LRA window provides audible/tactile transmission — vibration felt clearly on lid exterior
- 40. Confirm strap slots accept 22 mm NATO strap without forcing
- 41. Confirm enclosure does not flex or creak under normal wrist-worn loading
- 42. Confirm no wires pinched in lid seam — inspect perimeter with light before final close

2.7 Troubleshooting Quick Reference

#	Problem	Likely Cause	Solution
1	DRV2605L not found on I2C	SDA/SCL swapped; missing power;	Check wiring against pin table; verify 3.3

#	Problem	Likely Cause	Solution
		missing pull-ups	V on VIN; add 4.7 kΩ pull-ups; use explicit <code>Wire.begin(SDA, SCL)</code> in firmware
2	LRA doesn't vibrate	Wrong OUT pad polarity; auto-cal failure; no contact	Verify OUT+/OUT- leads; check auto-cal serial output; test manual RTP amplitude 200
3	Servo jitters or doesn't move	Insufficient voltage; wrong PWM pin; power supply noise	Verify 5 V on servo VCC; confirm GPIO5 in firmware; add 100 μF cap across servo VCC/GND
4	Wi-Fi won't connect	Wrong credentials; 5 GHz network; antenna obstructed	Check SSID/password case sensitivity; confirm 2.4 GHz band; clear antenna path on XIAO edge
5	Low battery not detected	LBO pin not read in firmware; wiring error; or GPIO2 not configured as INPUT_PULLUP	Read GPIO2 as digital input (active LOW below 3.2 V). Check PowerBoost LBO wiring to GPIO2. No divider required — simpler than ADC approach.
6	Watchdog resets loop	Motif duration total exceeds 8 seconds	Reduce step count or <code>dur_ms</code> values; split long motif into shorter sequential motifs
7	Enclosure too tight / won't close	Printer dimensional accuracy; warping; over-infill	Increase CLEARANCE variable in OpenSCAD by 0.5 mm; re-print; sand mating surfaces lightly
8	LRA too quiet through lid	Lid membrane too thick; LRA not fully seated	Enlarge LRA window in OpenSCAD; reduce <code>LID_T</code> parameter to 1.0 mm; press-fit LRA more firmly; add thin foam ring seal

2.8 Future Enhancements Roadmap

#	Enhancement	Description
1	BLE MIDI Support	Map MIDI note-on/off to LRA effects and servo positions in real time. Enables direct integration with DAWs and MIDI controllers for live performance haptics.
2	Multi-Node Mesh Networking	ESP-NOW P2P mesh for full-body haptic suit, sub-10 ms sync across wrist/chest/ankle nodes. No router required — direct peer addressing.
3	IMU Integration	MPU-6050 6-axis IMU for motion-reactive haptics and gesture-triggered feedback. Mount on internal perfboard, I2C address 0x68/0x69.
4	Custom PCB	Purpose-built 25 mm circular PCB integrating DRV2605L, voltage divider, servo header, LiPo connector, and optional IMU footprint. Eliminates perfboard and reduces assembly time.
5	OTA Firmware Updates	ArduinoOTA over-the-air — no need to open enclosure or remove strap for firmware changes. Triggered via UDP command or dedicated BLE characteristic.
6	Web-Based Motif Designer	Local web UI (AP mode + HTTP server) with drag-and-drop timeline editor for motif composition. Exports JSON motif files compatible with existing UDP schema.

Section 3 — Sensor Species Family — All Morphologies

3.0 Family Overview

Parameter	Value
Total primary components	20 (6 Pollen Pods + 4 Perch Pods + 6 Reef Plates + ~48 Garland Links + 8 bloom nodes + 16 drift fins)
Total estimated budget	~\$2,604 CAD
Build timeline	4 months (Phases 1–6)
Central compute (hub)	Raspberry Pi 4B — coordination layer, sensor aggregation, LED choreography broadcast
Node compute	Raspberry Pi Pico W (one per pod cluster) — I2C sensor polling, local LED data, Wi-Fi uplink to hub
Inter-pod comms	Wi-Fi to RPi 4B hub; I2C daisy-chain via JST-SH for sensors within each pod cluster
Power architecture	Central 5 V / 10 A switching PSU with per-branch fusing; individual Pico W LiPo backup cells for graceful failover

3.1 Morphology A — Pollen Pod (Sensor Capsule)

Parameter	Value
Quantity	6
Primary material	PETG (shell halves) + Clear PETG (optical elements)
Flexible element	TPU 95A lattice vent ring (equator — parametric spore-derived pattern)
Lighting	WS2812B LED ring (individually addressable, warm glow, sensor-reactive hue/intensity)
Form factor	Ovoid two-part clamshell, ~80–100 mm long (estimated)
Node MCU	Raspberry Pi Pico W

Description: Ovoid two-part clamshell. The TPU 95A lattice vent ring at the equator provides a parametric pattern derived from natural spore structures, allowing airflow to internal sensors while adding visual texture and organic identity. The WS2812B LED ring creates warm directional glow that shifts hue and intensity in response to sensor readings. Six pods are individually addressable and uniquely positioned in the garland network.

Sensor Payload:

Sensor	IC	Measurement	I2C Address	Notes
Temp/ Humidity/ Pressure	BME280	Temperature, relative humidity, barometric pressure	0x76 or 0x77	Address set by SDO pin; allows two per bus segment
Light (HDR)	TSL2591	0.1 to 88,000 lux high dynamic range	0x29	Fixed address — only one per I2C segment
Capacitive touch	TBD — MPR121 or single-pad AT42QT	Touch/proximity	TBD	MPR121 at 0x5A — CONFLICT with DRV2605L if combined on shared bus; assign carefully
LED ring	WS2812B	RGB output, individually addressable	Single-wire data	Not I2C — data line from Pico W GPIO

Print Specification: PETG shell (both halves) + Clear PETG for optical/lens elements + TPU 95A vent ring. All three materials required per pod. See Section 4.1 for material parameters.

3.2 Morphology B — Perch Pod (Sentinel Platform)

Parameter	Value
Quantity	4
Primary material	PETG (body)
Canopy	Clear PETG dome (optical diffuser)
Lighting	NeoPixel Jewel (7-LED array, lantern effect)

Parameter	Value
Form factor	Nest-form, larger than Pollen Pod, outward-facing anchor points
Node MCU	Raspberry Pi Pico W

Description: Larger, nest-like, outward-facing sentinel form. Four Perch Pods placed at key anchor points in the installation establish spatial hierarchy. The Clear PETG dome canopy acts as an optical diffuser over the NeoPixel Jewel — producing a lantern effect visible from distance that responds to viewer presence. The PIR motion sensor triggers proximity awareness; the APDS-9960 provides fine-grained gesture and proximity resolution and color sensing.

Sensor Payload:

Sensor	IC	Measurement	I2C Address	Notes
Motion (PIR)	PIR sensor module	Passive infrared motion detection	Digital GPIO	Not I2C; digital HIGH/LOW output to Pico W GPIO pin
Gesture / Proximity / Color	APDS-9960	Gesture direction, proximity distance, RGB+clear color	0x39	Fixed address — one per I2C segment
LED array	NeoPixel Jewel	RGB output, 7-LED array, diffused lantern	Single-wire data	Under Clear PETG dome diffuser

Print Specification: PETG body + Clear PETG dome canopy. Dome must be printed with dry Clear PETG to avoid bubble haze — see Section 4.1.

3.3 Morphology C — Reef Plate (Structural Substrate)

Parameter	Value
Quantity	6 panels
Dimensions	200 × 150 mm per panel
Primary material	PETG (indoor / gallery)
UV-stable variant	ASA (outdoor / UV-exposed / window-

Parameter	Value
	adjacent installations)
MCU	None — passive structural element
Mounting	Adjustable standoffs + snap-fit edge trim; modular configuration

Description: Textured structural backplanes referencing coral reef architecture — complex, layered, simultaneously rigid and richly detailed. Each panel exhibits a parametric surface texture maintaining family resemblance across the set while remaining individually distinct. Snap-fit edge trim and adjustable standoffs allow modular reconfiguration for different installation geometries. Provides physical mounting substrate for pods and visual backdrop unifying the overall composition.

Material Variants:

Variant	Material	Application	Special Requirements
Standard	PETG	Indoor / gallery installations	Standard print setup
UV-Stable	ASA	Outdoor / UV-exposed / window-adjacent	Enclosed print environment required to prevent warping

Print Specification: PETG or ASA; no flexible elements. No MCU, no electronics — structural print only. ASA requires enclosed chamber; see Section 4.2.

3.4 Morphology D — Garland Links & Kinetic System

Parameter	Value
Links (chain)	~48 articulating chain-link forms
Bloom nodes	8 branching hubs
Drift fins	16 (PETG-CF — fine, strong, slightly translucent)
Link material	PETG
Spacer rings	TPU 85A (softer articulation) or TPU 95A (firmer)
Drift fin material	PETG-CF — requires hardened steel nozzle

Parameter	Value
	+ all-metal hotend
Bloom node material	PETG
Rigging	Spectra/Dyneema braided tendon line
Lighting	WS2812B LED strip woven through garland channels

Description: Articulating chain-link forms connecting pods and plates into organic garlands spanning the installation volume. Bloom nodes serve as branching hubs at key junctions. Drift fins (PETG-CF — fine and slightly translucent) provide visual balance and kinetic flutter under air movement. TPU spacer rings between links allow natural low-resistance articulation — kelp-in-current quality. WS2812B LED strips woven through garland channels carry light pod-to-pod, sharing the same physical path as the Spectra/Dyneema tendon.

Tendon System: Spectra/Dyneema braided line (high tensile strength, minimal stretch, fine diameter) routes through printed cleats and spring-loaded tensioners. Servo-driven or solenoid actuators at key nodes translate sensor triggers (from Pollen and Perch Pods) into garland gestures. Spring tensioners return the system to neutral state. Light and motion share the same channel — gesture propagates along the same path as illumination.

Print Specification:

Component	Material	Critical Requirements
Chain links	PETG	Standard setup
Spacer rings	TPU 85A (soft) or TPU 95A (firm)	Direct drive extruder; drybox required
Drift fins	PETG-CF	Hardened steel nozzle MANDATORY; all-metal hotend required; drybox required
Bloom nodes	PETG	Standard setup

Section 4 — Fabrication Specifications

4.1 Materials Reference Table

Material	Role	Print Temp (est.)	Special Requirements	Notes
PETG	Primary structural (all morphologies)	230–250°C	Standard setup	Dimensionally stable, good layer adhesion, captures detailed surface texture well
TPU 95A	Pollen Pod vent ring; firmer flex elements	220–235°C	Direct drive extruder strongly preferred	Firm flex; parametric lattice rings; abrasion-resistant
TPU 85A	Garland spacer rings; softer articulation elements	215–230°C	Direct drive extruder mandatory; drybox required	Softer flex; kelp-quality articulation; moisture-sensitive — dry before and during print
PETG-CF	Drift fins; high-strength fine elements	240–260°C	Hardened steel nozzle MANDATORY; all-metal hotend; drybox required	Fine, strong, slightly translucent surface quality; highly abrasive — destroys brass nozzles immediately
ASA	Reef Plate UV-stable variant	240–260°C	Enclosed print environment required	UV and weather resistant; suited for outdoor / window-adjacent installations; warps severely without enclosure
Clear PETG	Optical diffusers — Pollen Pod lens, Perch Pod canopy dome	230–250°C	Dry filament critical — moisture causes bubbles and haze in optical sections	Excellent light transmission; printable optical elements; must be stored and printed dry

Material	Role	Print Temp (est.)	Special Requirements	Notes
PLA	Wrist-Puck enclosure (consumer-grade / prototype builds)	200–220°C	Standard setup	Stiffer and easier to print than PETG; less heat-resistant; suitable for indoor wearable use

4.2 Printer: Bambu Lab A1 Mini (In Hand)

The Bambu Lab A1 Mini is the primary production printer for this project. It is already owned and partially equipped. The table below reflects its actual current status — not a wishlist.

Requirement	A1 Mini Status	Action Required
Direct drive extruder	✓ Built-in	None — TPU 95A and 85A are straightforward on this machine.
Hardened steel nozzle	✗ Not yet acquired	Purchase Bambu hardened nozzle accessory before any PETG-CF print run. Mandatory — PETG-CF destroys brass nozzles.
All-metal hotend	✓ Likely qualifies	Confirm max temp spec against ASA requirement (~260°C). A1 Mini hotend is rated to 300°C — should be sufficient.
Enclosed print environment	✗ Not enclosed	Gap for ASA Reef Plate variants. Options: (a) Bambu enclosure tent accessory, (b) draft shield in slicer + still-air environment, (c) print ASA only during calm, warm conditions. PETG and TPU are unaffected — open frame is fine for both.
Humidity / drybox control	External solution required	TPU and PETG-CF must be dried before and during printing. Use a filament dryer or sealed container with desiccant. A1 Mini AMS Lite can assist with dry storage for standard

Requirement	A1 Mini Status	Action Required
		filaments but is not rated for TPU — run TPU externally.
Layer height / infill	✓ Full slicer control	0.2 mm / 20% infill as specified. Use Bambu Studio profiles.

Net Status — Mostly Ready

Two items to resolve before full campaign:

1. **Acquire hardened nozzle** (Bambu accessory, ~\$15-25 CAD) — blocks PETG-CF drift fin printing.
2. **Decide ASA strategy** (enclosure tent or workaround) — blocks Reef Plate outdoor-variant printing only. Indoor PETG Reef Plates can proceed immediately.

Everything else — all Pollen Pod shells, Perch Pod bodies, TPU vent rings and spacer rings, Garland Links and bloom nodes, Wrist-Puck enclosures — can be printed now.

4.3 Assembly Hardware

Component	Specification	Application
Heat-set brass inserts	M2 and M3 (specify per design file)	All threaded connections in printed parts — permanent, reliable
JST-SH connectors	1.0 mm pitch, hand-crimped	I2C daisy-chain wiring, sensor interconnects within pod clusters
JST-PH connectors	2.0 mm pitch	Battery and power connections (LiPo, PSU branch lines)
Spectra/Dyneema line	Fine braided, high tensile strength, minimal stretch, fine diameter	Tendon rigging throughout garland kinetic system
Spring tensioners	TBD by design — size to	Garland kinetic return-to-

Component	Specification	Application
	garland run length and ceiling height	neutral on sensor trigger release
M2 hardware	M2×6 mm screws + M2 nuts (nylon preferred to avoid shorts)	Board mounting in Wrist-Puck enclosure
Hookup wire	26–30 AWG silicone stranded, assorted colors	All inter-board wiring in both platforms

4.4 Tools Required

Tool	Specification
Soldering iron	Fine tip, 300–350°C
Solder	Lead-free, 0.5 mm preferred for fine work
Wire strippers	30 AWG capable
Flush cutters	Lead trimming and fine wire
Small Phillips screwdriver	M2 size
3D printer	FDM, 0.4 mm nozzle standard; hardened steel nozzle for PETG-CF production runs
Multimeter	Continuity, voltage, and resistance modes
Needle-nose pliers	Linkage wire bending; JST housing manipulation
Crimping tool	JST-SH and JST-PH crimp dies — proper ratchet crimper preferred
Hot glue gun	Optional — LiPo cell securing inside enclosures
Heat-set insert tool	M2/M3 heated tip — fits soldering iron handle
Computer	Arduino IDE 2.x (Wrist-Puck firmware); Python / MicroPython environment (Pico W nodes)

Section 5 — Electronics & Compute Reference

5.1 Wrist-Puck Electronics Stack

Component	IC / Module	Interface	I2C Address	Notes
MCU	Seeed XIAO ESP32-C3	—	—	Wi-Fi 802.11 b/g/n + BLE 5.0; USB-C; built-in LiPo charge; 21×17.8 mm footprint
Haptic driver	DRV2605L	I2C	0x5A	123-effect waveform library + RTP mode; auto-calibration for LRA resonant frequency
LRA	Coin LRA C10-100, ~175 Hz	Via DRV2605L	—	DRV2605L drives LRA directly via OUT+ / OUT-pads
Servo	Adafruit SG51R	PWM — GPIO5	—	Range 750–2250 μ s; center 1500 μ s; 0.6 kg·cm at 4.8 V
Power management	Adafruit PowerBoost 500C	USB-C + BAT	—	Boost 3.7 V LiPo → 5.2 V regulated; 500 mA charge; LBO pin at 3.2 V threshold. In hand.

5.2 Sensor Species Family Electronics Stack

Component	IC / Module	Interface	I2C Address	Notes
Hub MCU	Raspberry Pi 4B	Ethernet / Wi-Fi	—	Central coordination, sensor data aggregation, LED choreography broadcast to all nodes
Node MCU	Raspberry Pi	Wi-Fi to hub	—	One per pod

Component	IC / Module	Interface	I2C Address	Notes
	Pico W (×6+)			cluster; drives local I2C sensors and LED data lines; Wi-Fi uplink to RPi 4B hub
Temp/ humidity/ pressure	BME280	I2C	0x76 or 0x77	Pollen Pods only; SDO pin sets address — allows two per bus segment if needed
Light (HDR)	TSL2591	I2C	0x29	Pollen Pods only; range 0.1–88,000 lux; accommodates Yukon midnight sun through polar night
Gesture / proximity / color	APDS-9960	I2C	0x39	Perch Pods only; gesture direction, proximity, RGB+clear color sensing
Motion (PIR)	PIR sensor module	Digital GPIO	—	Perch Pods only; passive infrared; digital HIGH/LOW output to Pico W GPIO
Capacitive touch	TBD — MPR121 or similar	I2C	TBD	Pollen Pods only; MPR121 at 0x5A — conflicts with DRV2605L if on shared bus
LED (pods)	WS2812B ring	Single-wire data	—	Individually addressable; per-pod, driven from Pico W GPIO
LED (garland)	WS2812B strip	Single-wire data	—	Woven through garland channels; shared data line with pod node
LED (perch)	NeoPixel Jewel	Single-wire data	—	7-LED array under Clear PETG dome

Component	IC / Module	Interface	I2C Address	Notes
				diffuser; lantern effect
Power supply	5 V / 10 A switching PSU	—	—	Central feed; per-branch fusing; size to final LED count (60 mA/LED at full white)
LiPo backup	Per-node Pico W LiPo cell	—	—	Graceful failover on PSU branch fault; holds last LED state; recharges from PSU branch

5.3 Power Architecture Comparison

Platform	Primary Power	Backup	Max Draw (est.)	Charge Method
Wrist-Puck	3.7 V LiPo → PowerBoost 500C → 5.2 V regulated	N/A (PowerBoost handles charge + boost)	~250 mA avg (servo active, 5.2 V rail)	USB-C via PowerBoost (500 mA, single board)
Sensor Species	5 V / 10 A central switching PSU	Per-node LiPo (Pico W); holds last LED state on branch fault	~2 A total est. (20 LEDs active + sensors + node MCUs)	Per-node USB when operating in isolated / bench mode

PSU Sizing Note

WS2812B LEDs draw up to 60 mA per LED at full white (all channels max). For the garland system, calculate total LED count × 60 mA and add 20% headroom. The 5 V / 10 A PSU (50 W) is sized conservatively for the v1 build — verify against final LED count before installation.

Section 6 — Sensor Species Family Budget (Full Breakdown)

6.1 Budget Summary Table

Category	Estimated Cost (CAD)
Morphology Components — 3D Printed Parts (material cost)	\$320
Filaments — PETG, TPU 95A/85A, PETG-CF, ASA, Clear PETG	\$325
Printer Upgrades & Fabrication Equipment	\$320
Mechanical Hardware — fasteners, shafts, springs, heat-set inserts	\$195
Wire, Cable & Tendons — Spectra/Dyneema line, hookup wire	\$104
Electronics — Sensors (BME280, TSL2591, APDS-9960, PIR, capacitive)	\$285
Electronics — Controllers & Computing (Pico W nodes, RPi 4B hub)	\$318
Electronics — Power (PSU, LiPo cells, fusing, distribution hardware)	\$215
Electronics — Lighting & Effects (WS2812B strips, NeoPixel Jewels, LED rings)	\$122
Connectors & Wiring Accessories (JST-SH, crimps, heat shrink, cable management)	\$185
Consumables & Workshop Supplies (solder, flux, adhesives, isopropanol, tooling)	\$215
Grand Total (Estimated)	~\$2,604 CAD

Whitehorse Shipping Surcharge

Shipping to Whitehorse, YT may add **10-15%** to electronics and specialty component costs (DigiKey, Mouser, Adafruit, etc.). Treat as contingency in project budget planning. Order in consolidated batches where possible to minimize per-shipment overhead.

Full Itemized BOM

Full 103-item itemized BOM available in supporting spreadsheet: **Sensor Species Family — Full Bill of Materials**. This section provides the category-level summary only. Refer to supporting spreadsheet for per-line SKU, supplier, quantity, and unit pricing.

Section 7 — Project Phases & Timeline (Sensor Species Family)

Phase 1 — Printer Upgrades & Material Procurement | Month 1

Acquire and install hardened steel nozzle, all-metal hotend, and direct drive extruder upgrade. Print or source printer enclosure. Procure full filament stock across all five material types (PETG, TPU 95A, TPU 85A, PETG-CF, ASA, Clear PETG). Establish drybox humidity-controlled workflow for TPU and PETG-CF storage and active printing. Run material-specific calibration prints to verify temperature, retraction, and dimensional accuracy across all materials before launching main print campaign. Document any printer-specific tuning settings per material.

Phase 2 — Print Campaign | Months 1-2

Print all morphology components in order: Pollen Pod shells and vent rings, Perch Pod bodies and Clear PETG canopy diffusers, Reef Plates (all 6 panels — PETG and/or ASA variants as applicable), full Garland Link set including bloom nodes and drift fins. Quality-check each part family against fit and finish standards before proceeding to next morphology. Batch printing by material type to minimize nozzle changes and minimize drybox transitions. Tag and store completed parts by morphology and position designation.

Phase 3 — Electronics Assembly | Month 2

Solder and assemble all sensor breakout boards. Assign and verify I2C addresses for each pod cluster — check for conflicts per Section 8.3 address table. Wire and test Pico W node circuit boards individually before integration. Assemble and bench-test all LED chains (WS2812B rings, strips, NeoPixel Jewels). Establish power distribution wiring harnesses with per-branch fusing. Bench-test all sensor and LED subsystems before integration into printed shells. Document any I2C bus issues or address conflicts encountered.

Phase 4 — Rigging & Integration | Month 3

Hand-rig Spectra/Dyneema tendon system through assembled garland chain. Mount and assemble garland links into complete runs. Integrate electronics into pod shells using heat-set inserts and hand-crimped JST-SH connectors. Route and dress all cabling neatly through garland channels — WS2812B data and tendon share the same channel. Mount Reef Plates with adjustable standoffs and attach garland assemblies to substrate. Verify full range of garland motion and confirm tendon routing does not restrict articulation.

Phase 5 — Calibration & Software | Month 3

Configure sensor firmware on all Pico W nodes: I2C addressing, polling rates, Wi-Fi credentials, data format to hub. Set up RPi 4B hub: data aggregation scripts, sensor-to-response mapping, LED choreography broadcast. Calibrate LED response curves to sensor input ranges (TSL2591 lux → LED hue, BME280 humidity → intensity, etc.). Tune tendon tension and spring pre-load to venue ceiling height and garland run lengths. Conduct full system integration test with all pods, garland, and hub active simultaneously.

Phase 6 — Installation, Documentation & Public Presentation | Month 4

Full installation in target venue. Final in-situ calibration of sensor response curves and tendon tension. Professional photography and video documentation of installed

system (for funder reporting and portfolio). Public presentation event. Preparation of complete documentation package: assembly notes, calibration settings, wiring diagrams, and design evolution log entries. Confirm all systems operational before handoff.

Section 8 — Deployment Considerations

8.1 Wrist-Puck Deployment

- **Network:** 2.4 GHz Wi-Fi only — ESP32-C3 does not support 5 GHz. Static IP recommended for reliable UDP delivery; DHCP lease changes will break UDP sender targeting.
- **Multi-node:** For full-body haptic suit, use ESP-NOW peer mesh (Enhancement #2). Current v1.0 is single-node UDP only.
- **Latency:** UDP motifs delivered fire-and-forget; no acknowledgement. For time-critical multi-node sync, add sequence IDs to motif schema and implement receiver-side dedup.
- **Wearability:** ~55 g target weight. LiPo must not be compressed inside enclosure — creates internal pressure and thermal risk. Strap must allow full circulation — test fit for at least 5 minutes before extended wear session.
- **Safety — idle timeout:** Firmware auto-centers servo after 5 seconds of no motifs received. Prevents servo hold current drain and mechanical stress on sliding mass rail.
- **Safety — hardware watchdog:** Resets on loop block exceeding 8 seconds. Any motif step sequence totalling >8 s will trigger reset — design motifs accordingly (see Section 2.7, item #6).
- **Power:** PowerBoost 500C is the single charge and boost layer. LBO pin triggers firmware idle-and-alert at 3.2 V (cleaner than ADC divider). Servo

runs at 5.2 V on battery — full rated torque maintained throughout LiPo discharge curve.

- **Firmware update:** Requires USB-C cable and physical enclosure opening in v1.0. OTA update planned as Enhancement #5.

8.2 Sensor Species Family Deployment

- **Venue power:** 5 V / 10 A PSU on a single circuit with per-branch fusing. Size PSU to final LED count: WS2812B draws up to 60 mA per LED at full white. Protect each branch independently — a single LED strip short must not take down the whole installation.
- **Yukon environmental conditions:** TSL2591 range (0.1–88,000 lux) accommodates subarctic seasonal extremes — Whitehorse midnight sun through polar night. BME280 humidity and temperature sensors calibrated for interior continental climate (low humidity, wide temperature range).
- **Indoor vs. outdoor variants:** PETG panels for indoor / gallery. ASA panels for outdoor or UV-exposed / window-adjacent. All electronics require indoor or fully weather-protected enclosure regardless of panel material.
- **Garland rigging:** Hand-rig on-site. Spring pre-load tuned to venue ceiling height and garland run length — what works in the studio may need adjustment on-site. Budget time for on-site rigging calibration in Phase 6.
- **Graceful failover:** Per-node LiPo backup holds last LED state on PSU branch fault. Hub automatically reconnects nodes on Wi-Fi restore — no manual intervention required for temporary network interruption.
- **Scalability:** Morphological framework designed to grow. Additional Pollen or Perch Pods can be added to I2C bus (up to address limits per segment). New Pico W nodes added to hub Wi-Fi mesh. New garland runs attached to existing bloom nodes.
- **Documentation:** Full 103-item BOM available as supporting spreadsheet. Professional photography/video planned in Phase 6 for funder reporting.

8.3 Cross-Platform Notes

- **JST connector vocabulary:** Both platforms use JST-SH (1.0 mm pitch) for I2C daisy-chain wiring and JST-PH (2.0 mm pitch) for power. Standardize on these across all future variants to maintain tooling compatibility and reduce crimping errors.
- **WS2812B as common LED language:** All current and future variants should maintain WS2812B compatibility for unified LED choreography control from the RPi 4B hub or equivalent central controller.
- **Whitehorse shipping:** Both platforms are designed for Whitehorse, YT procurement realities — 10–15% Whitehorse shipping surcharge on specialty components from Mouser, DigiKey, Adafruit. Order in bulk where possible; treat surcharge as budget contingency.
- **I2C address conflict table:**

Device	I2C Address	Conflict Notes
DRV2605L (Wrist-Puck haptic driver)	0x5A	Fixed — no alternate address available
BME280 (Pollen Pod)	0x76 or 0x77	SDO pin selects — allows two per bus segment
TSL2591 (Pollen Pod)	0x29	Fixed — one per I2C segment
APDS-9960 (Perch Pod)	0x39	Fixed — one per I2C segment
MPR121 (Pollen Pod capacitive touch — TBD)	0x5A (default)	CONFLICT with DRV2605L. MPR121 addresses can be shifted (0x5A/5B/5C/5D via ADDR pin) — but DRV2605L is fixed at 0x5A. If both appear on a shared bus, MPR121 must be addressed to 0x5B, 0x5C, or 0x5D. Verify before bus design is finalized.

- **Mechanical wire routing:** Both platforms use Spectra/Dyneema (garland tendons) or silicone-jacketed wire (wearable) for mechanical routing — preferred for flexibility and longevity in articulated structures. Standard PVC-jacketed wire not recommended for wearable or garland applications.

Section 9 — Design Evolution Log

Living log. Append new entries as design iterations progress. Each entry should capture: platform, what changed (or was established as baseline), and date. Do not overwrite previous entries.

Version / Iteration	Platform	Change / Note	Date
Wrist-Puck v1.0	Wrist-Puck	Initial design. Dual-channel haptic: DRV2605L LRA (texture/vibrotactile) + SG51R servo with sliding brass mass (weight-shift). MCU: Seeed XIAO ESP32-C3. Protocol: UDP port 9000, JSON motifs. Canada-sourced BOM ~\$115–135 CAD per node. 14-item BOM, all suppliers confirmed May 2026.	May 2026
Sensor Species v1.0	All four morphologies (Pollen Pod / Perch Pod / Reef Plate / Garland Links)	Initial build campaign. 20 primary components. Four-morphology family: A=Pollen Pod (×6), B=Perch Pod (×4), C=Reef Plate (×6), D=Garland Links (~48 links + 8 bloom nodes + 16 drift fins). Node compute: RPi Pico W per cluster. Hub: RPi 4B. Architecture:	May 2026

Version / Iteration	Platform	Change / Note	Date
		distributed I2C within clusters + Wi-Fi uplink to hub. Full material set: PETG/TPU 95A/TPU 85A/PETG-CF/ASA/Clear PETG. Total budget: ~\$2,604 CAD. 4-month build timeline.	
Wrist-Puck v1.0 Rev A	Wrist-Puck	Power architecture revised: Adafruit PowerBoost 500C (in hand) replaces TP4056 + 100 kΩ voltage divider. Single-board charge + boost → 5.2 V rail. Servo now runs at rated voltage on battery. LBO pin replaces ADC battery monitor. Printer confirmed as Bambu Lab A1 Mini (owned); hardened nozzle and ASA enclosure strategy still pending.	May 2026
<i>[Future entries — append here]</i>			

Section 10 — Quick Reference Card

Rapid-lookup summary. Confirm against full sections for assembly or deployment decisions.

Wrist-Puck v1.0 — At a Glance

MCU

Seeed XIAO ESP32-C3 (21×17.8 mm, Wi-Fi + BLE 5.0, USB-C, built-in LiPo charge)

Haptic Ch. 1

DRV2605L (I2C 0x5A) + 10 mm LRA C10-100 ~175 Hz (texture / vibrotactile)

Haptic Ch. 2

Adafruit SG51R servo (PWM GPIO5, 750–2250 μ s, center 1500 μ s) + brass sliding mass ~8–10 g (weight-shift)

Comms

UDP port 9000, JSON motifs, 2.4 GHz Wi-Fi only

Battery

3.7 V 500 mAh LiPo, USB-C charge | Runtime: 2–4 hr | Low-batt cutoff: 3.3 V

Power Mgmt

Adafruit PowerBoost 500C (in hand) — LiPo → 5.2 V regulated · LBO pin → GPIO2 (active LOW at 3.2 V)

Pin Map

I2C SDA=GPIO6 · SCL=GPIO7 · Servo PWM=GPIO5 · LBO (low-batt)=GPIO2

Enclosure

68 mm dia × 22 mm | PLA or PETG | 0.2 mm / 20% infill | 22 mm strap slots |

Printer: Bambu A1 Mini

BOM Cost

~\$115–135 CAD per node (PowerBoost 500C in hand; Canada-sourced, May

2026)

Sensor Species Family — At a Glance

Hub

Raspberry Pi 4B — coordination, aggregation, LED choreography broadcast

Nodes

Raspberry Pi Pico W — one per pod cluster; I2C sensors + LED data + Wi-Fi uplink

Sensors

BME280 (0x76/77) · TSL2591 (0x29) · APDS-9960 (0x39) · PIR (GPIO) ·

Capacitive TBD

LEDs

WS2812B rings (Pollen Pods) + WS2812B strip (garland) + NeoPixel Jewel (Perch Pods)

Power

5 V / 10 A central PSU + per-node LiPo backup (graceful failover)

Components

6 Pollen Pods (A) + 4 Perch Pods (B) + 6 Reef Plates (C) + ~48 garland links + 8 bloom nodes + 16 drift fins (D)

Filaments

PETG · TPU 95A · TPU 85A · PETG-CF (hardened nozzle req.) · ASA (enclosure req.) · Clear PETG

Budget

~\$2,604 CAD total | 4-month build timeline | +10–15% Whitehorse shipping contingency

Key I2C Address Map — Both Platforms

Device

Platform

I2C Address

Conflict

DRV2605L haptic driver

Wrist-Puck

0x5A

Conflicts with MPR121 default

BME280 temp/humidity/pressure

Sensor Species (Pollen)

0x76 / 0x77

None (two addresses available)

TSL2591 HDR light

Sensor Species (Pollen)

0x29

None

APDS-9960 gesture/proximity

Sensor Species (Perch)

0x39

None

MPR121 capacitive touch (TBD)

Sensor Species (Pollen)

0x5A (default) — CONFLICT

Must shift to 0x5B/5C/5D via ADDR pin if coexisting with DRV2605L

Haptic Pucks & Pollen Pods — Complete Variant Reference | Rev 1.0 | May 2026

Sean Pond, Whitehorse, YT, Canada | Living document — append Design Evolution
Log (Section 9) as iterations progress.